

Oyon, Md. Shihab Sharar

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Skills, Tools, Platforms & Competencies

Programming	Python, C, C++		
Designing	AutoCAD (2D, 3D, Electrical), SolidWorks, Autodesk Fusion 360		system, docker, container, kind from K8s), basic networking, device monitoring (IoT)
Simulation	Comsol Multiphysics, MATLAB, Simulink, Multisim, Homer Pro	AI, & ML	Computer Vision: YOLO based detection, data annotation on Roboflow Software, model training/testing, threshold tuning
Embedded	Arduino, ESP, Raspberry Pi, Jetson Orin Nano	Others	Data Analysis, Documentation (SOPs), workflow management, procurement support, Vendor coordination
Productivity	MS Office, Google Workspace (Docs/Sheets/Slides), LaTeX, github		
Systems	Linux fundamentals (bash scripting, file		

Experience

Dr. Anwarul Abedin Institute of Innovation (D2A2I) — American International University - Bangladesh (AIUB)

Research Assistant — Research and Development (R&D)

September 2024 - Present

- Coordinates day to day technical and administrative operations by guiding team workflows, balancing workloads, and ensuring timely completion of bill ready preparation, submissions, and purchase requests.
- Manages end to end procurement support: product requirement confirmation, vendor communication, quotation comparison, purchase processing, and delivery follow up to keep projects on schedule.
- Provides technical oversight and troubleshooting across hardware and software tasks, including IoT systems, computer vision applications, and AI driven solutions, ensuring smooth execution and minimizing operational delays.
- Supervises and offers solutions to critical problems for the project's equipment.
- Collaborates on research endeavors to reduce components, costs, and refine algorithms.
- Contributes significantly to the development of research papers that advance project objectives.
- Formulates technical reports and presentations on a weekly and monthly basis, incorporating data analysis and insights from project metrics.

Research Assistant (Intern) — Research and Development (R&D)

February 2024 – August 2024

- Examining diverse and big data from the TRP (Television Rating Point) measurement project to identify challenges and advancements
- Utilized Linux system to monitor TRP devices based on Raspberry Pi technology

Education

American International University - Bangladesh (AIUB)

Master of Science in Electrical and Electronic Engineering (MEEE) — CGPA 3.55 October 2024 – June 2026 (Defense Pending)

Bachelor of Science in Electrical and Electronic Engineering (EEE) — CGPA 3.77 September 2024

Lions School and College, Rangpur (LSCR) — HSC — GPA 4.75 2019

Police Line School and College, Rangpur (PLSCR) — SSC — GPA 4.82 2017

Awards & Certifications

- Dean's List Honors - For successfully achieving 3.95 in Fall 21-22 semester, 3.82 in Fall 22-23 semester, 3.8 in Spring 20-21 semester, and, 3.75 in Spring 21-22 semester
- Medal - For volunteering at AIUB Premier League 2023
- Oracle Cloud Infrastructure 2023 AI Certified Foundations Associate—Credential ID 307034687OCI23AIFCA—Dec-2023
- Corporate Etiquette — Credential ID 64a80b06692d1 — Jul-2023
- Spoken English — Credential ID 64a7d2127c6de — Jul-2023

Projects

Navigotech Project

Oct 2025 – Present

- Data collection, annotation on Roboflow Software for AI driven system of the Wheelchair.
- Designed and optimized 3D mechanical components (holders, enclosures, mounts) using Autodesk Fusion 360, ensuring practical assembly and field usability.
- Mentored and guided interns on 3D design best practices, design reviews, and implementation improvements to maintain quality and consistency.
- Performed hardware compatibility verification (interfaces, power, fitment) and supported procurement planning for required components.

- Executed hands-on integration work, including wiring, assembly, and validation; supported embedded coding and functional testing for deployed configurations.

Television Rating Point (TRP) Project

Feb 2024 – Present

- Updates and improves YOLO-based image processing models within the TRP system through dataset preparation, annotation coordination, training/testing experiments, and threshold tuning for better detection accuracy.
- Conducts parameter experimentation for training and evaluation, and resolved performance issues through systematic testing and optimization.
- Maintains and updates IoT-based device monitoring systems, ensuring stable connectivity, accurate telemetry, and system reliability.
- Coordinates with external vendors to diagnose issues, communicate technical requirements, track fixes, and ensure resolution of raised incidents within agreed timelines.

Cloud-Based Intelligent & Smart Ceiling Robots for Warehouse Management System

Feb 2023 – Jan 2024

- Developed a prototype and published a research paper on IEEE
- Submitted a second manuscript on a Q3 Journal (IJCDS)

Designing and Manufacturing a Single-Phase Transformer and Analyzing Its Performance

Jun 2022 – Nov 2022

- Developed a prototype and published a research paper on Journal of Energy Research and Reviews

Publications

Published

Masters - AIUB

1. Numerical Analysis of Balanced Geometry Selection in Circular PCF SPR Biosensors for Cancer and Bacterial Detection, Published on the Optics Continuum from Optica Publishing Group, USA, First Author.
2. Highly Sensitive Dual-Mode PCF-SPR Biosensor for rapid detection of PC-12 Adrenal Cancer cells, Presented on ICTP25 IEEE Conference, First Author.
3. Highly Sensitive Circular PCF based SPR Biosensor for Early Blood Cancer Detection, Presented at COMPASConf25 IEEE Conference, Corresponding Author.
4. Unlocking Early Detection: A Highly Sensitive Circular Shaped PCF-Based SPR Biosensor for Breast Cancer, Accepted on Springer Conference, Fourth Author.
5. Long Haul Optical Transmission Using Multi-Channel OAM-PDM Multiplexing over HMMF-FSO Communication System, Accepted on Springer Conference, Corresponding Author.

Research Assistant - D2A2I

1. Design and Optimization of an IR-Based Automation System with Telegram Bot Integration for Remote Device Control, Presented on ICREST25 (IEEE Conference), Corresponding Author.
2. Smart Parking System with Optical Character Reading Technology, Presented at ICCIT24 (IEEE Conference), Corresponding Author.
3. An IoT-Based Autonomous System for Oil Spill Detection, Article of Emerging Science Innovation, Corresponding Author.

Bachelor - AIUB

1. Design and Performance Analysis of Hydrogen Fuel Cell Vehicles in Bangladesh, Presented at PEEIACON24 (IEEE Conference), First Author.
2. Cloud-Based Ceiling Robot Assisted Intelligent Warehouse Management System, Presented at PEEIACON24 (IEEE Conference), First Author.
3. Modeling and Analysis of Small-Scale Solar PV and Li-ion Battery-based Smartgrid System, Presented at ICAEEE24 (IEEE Conference), Corresponding Author.
4. Integrating AE-CNN with Smart Relaying and SSCB for Enhanced Three-Phase Fault Detection and Mitigation, Presented at ICPS23 (IEEE Conference), Corresponding Author.
5. Three Phase Fault Analysis Using Thermal-Magnetic Circuit Breaker and Overcurrent Relay, Presented at ICICT4SD23 (IEEE Conference), First Author.
6. Designing and Manufacturing a Single-Phase Transformer and Analyzing Its Performance, Article of Journal of Energy Research and Reviews, Corresponding Author.

Submitted

1.

Numerical and ML Based Analysis of a Highly Sensitive PCF SPR Sensor for Wide Range Refractive Index Sensing and Biomedical Applications, Submitted on Arabian Journal for Science and Engineering (Review Phase), Third Author.
2.

Development of an Autonomous Logistics Robot for Modern Warehousing with Lifting and Navigation Capabilities, Submitted on IJCDs Q3 Journal, Corresponding Author.
3.

CNN and Transformer-Based Object Detection Integrated with Dijkstra’s Algorithm for Warehouse Navigation and Conflict Resolution, Submitted on IJCDs Q3 Journal, First Author.

Membership & Extra-Curricular Activities

The Institution of Engineers, Bangladesh (IEB), Associate Member, A- 27833	Aug 2025 – Present
Institute of Electrical and Electronics Engineers (IEEE), Graduate Member, 99787867	2024 – Present
Engineering Students Association of Bangladesh (ESAB), General Member	2023 – Present
AIUB Research for Development Club, R&D Advisor	Apr 2025 – Mar 2026
DUKE of Edinburgh Award Program (Direct Gold)	Jan 2022 – Mar 2025
AIUB FILM CLUB (General Member)	Sep 2022 – Nov 2024

References

Bachelor’s and Master’s Supervisor	Workplace Supervisor
Prof. Dr. Mohammad Nasir Uddin P.Eng	Dr. M. Tanseer Ali
Professor & Head of EEE Dept. (Graduate Program), FE	Associate Professor, Special Assistant [EEE], FE
American International University-Bangladesh	American International University-Bangladesh
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